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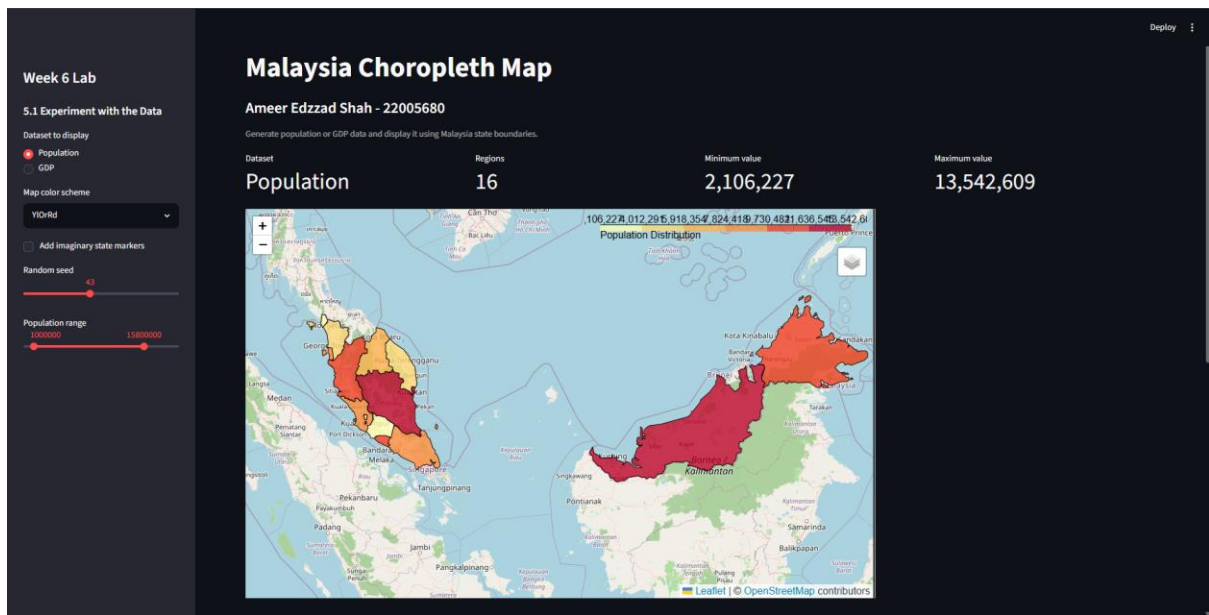
In this lab, I experimented with a Malaysia choropleth map by changing the generated data, adjusting the value range, switching the dataset type, and testing different color schemes. The original map displayed generated population values for each Malaysian state and federal territory. When the population range was changed to between 1 million and 10 million, the map showed clearer differences between regions. States with higher generated population values appeared in darker colors, while states with lower values appeared in lighter colors. This made it easier to compare which areas had larger or smaller populations.

I also changed the dataset from population to GDP. This affected the meaning of the map because the visualization no longer represented the number of people, but instead represented economic value. Although the map layout stayed the same, the color distribution changed because the generated GDP values were different from the population values. This showed that the same choropleth map can be reused for different types of numerical data as long as the data is linked to the correct state names.

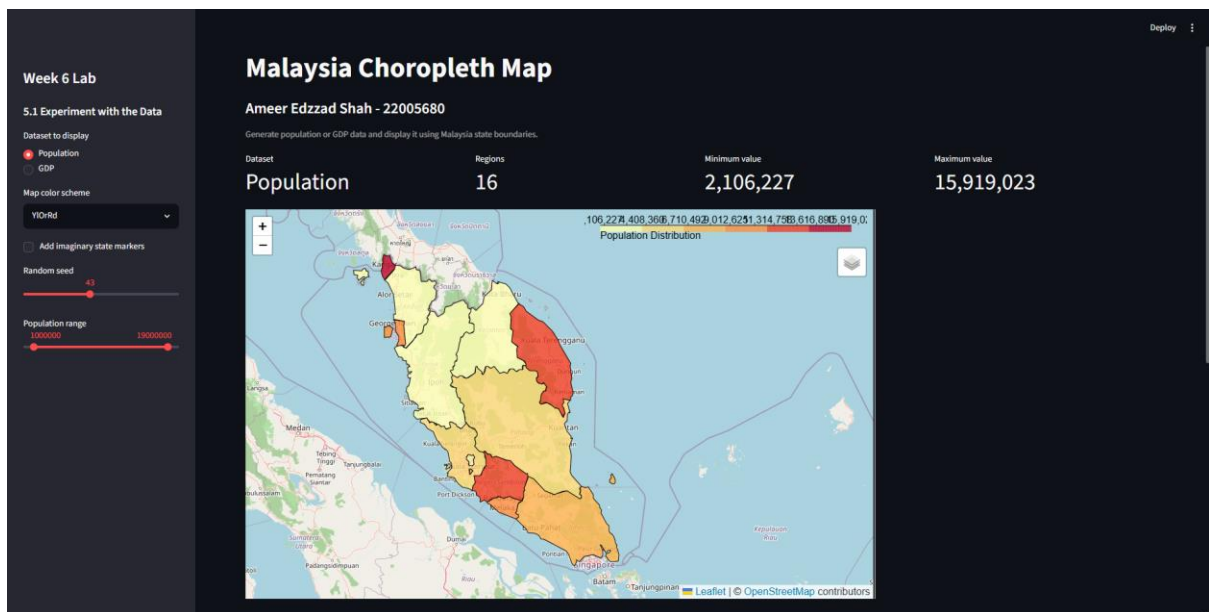
Changing the color scheme also affected how the map was interpreted. The YlOrRd color scheme made high values stand out strongly using red and orange tones. The Blues and Greens schemes created a calmer appearance, while BuGn gave a clear contrast between low and high values. These changes showed that color choice is important because it influences how quickly users can identify patterns.

Finally, I tested adding imaginary state markers. These markers did not become part of the choropleth regions because they were not included in the Malaysia GeoJSON boundary file. However, they appeared as point markers on the map. This showed that choropleth maps require proper geographic boundary data, while marker points only need latitude and longitude coordinates.

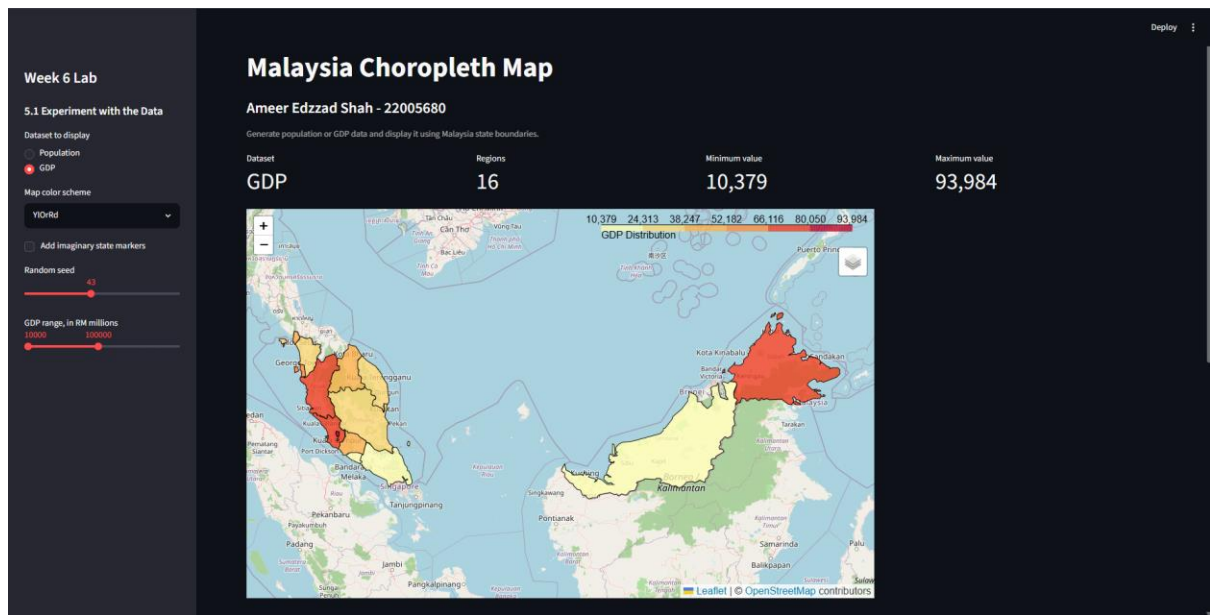
Population map using YlOrRd



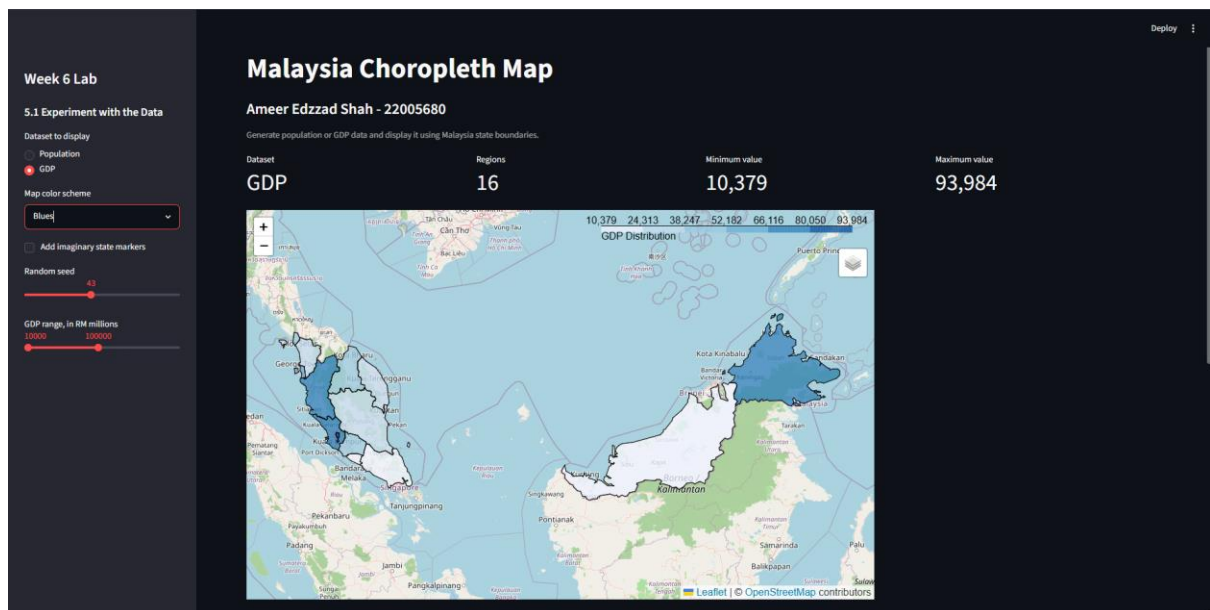
Population map with a changed population range

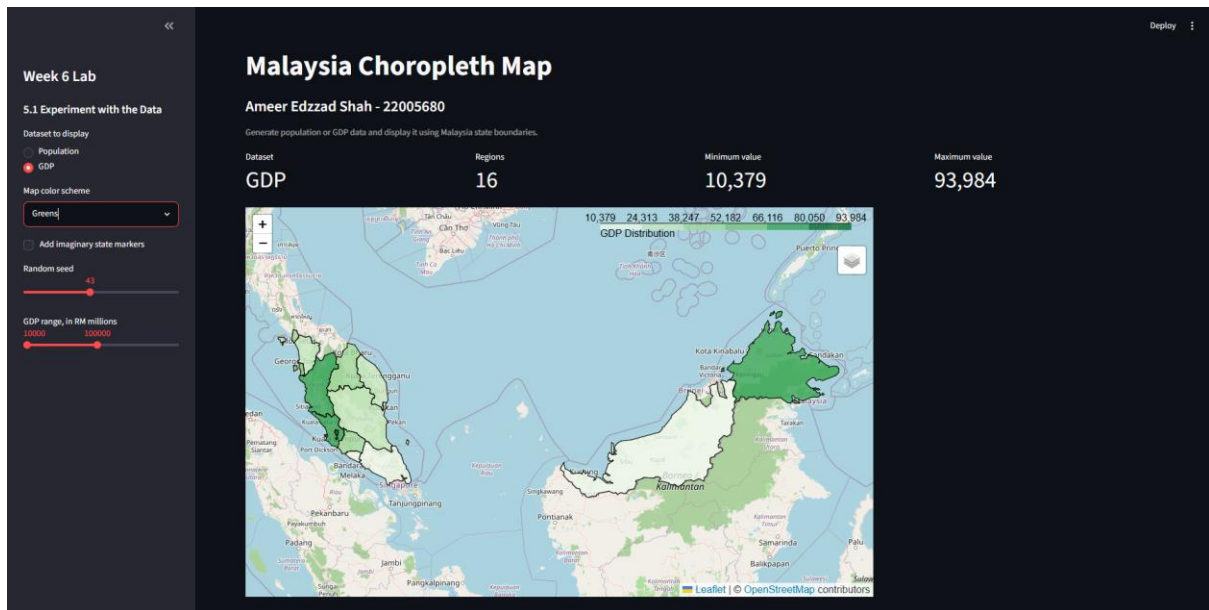


GDP map



Map using another color scheme such as Blues or Greens





Map with imaginary state markers enabled

